Image Processing-based Pothole Detecting System for Driving Environment

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***Abstract*—The pothole is one of the significant factors for the motorist. It can cause a car accident, the decrease of car lifetime and a decrease of the motorist concentration. There are three kinds of the pothole detecting methods, such as using a 3-axis acceleration sensor, a camera sensor and, a laser sensor. All of three have the cons and pros. Our proposed system is based on an image processing method with the camera sensor. Detecting the potholes based on the potholes characteristic. The characteristics of the potholes are the dark region, the round shape and the rugged texture.**

I. Introduction

The asphalt pavement condition is one of the significant factors for motorist [1]. Such as a pothole, crack, rut is factors of the asphalt pavement condition. Bad asphalt pavement condition can cause various problems such as a car accident, decrease of motorist concentration and the shock of car suspension. It is because when the wheel of the car falls in pothole can cause trouble to car suspension system or motorist concentration. As a result, the pothole is the main factor of which decrease car lifetime.

To repair the location of the pothole, an acquisition of its position must be required. But most of the city depends on motorist report. For this reason, an automatic pothole detecting system is needed. Therefore, three styles of the automatic pothole detect systems are suggested [2]. First, detect the pothole using 3-axis acceleration sensor in the car when the wheel of the car going on the pothole and analyzing the sensor reacting [3][4][5]. This method can detect the pothole fast and does not need much in initial cost. However, when the car does not go over the pothole, the system cannot be detected pothole. Second, detecting the pothole with a laser sensor [6]. By driving on the asphalt pavement, the laser sensor scans the asphalt pavement environment. The laser scan makes depth image of the asphalt pavement to find the potholes. It is the high accuracy of detecting the pothole. Nonetheless a laser scanning device required high-cost and its detecting speed is slower than other methods. So, detecting speed is the most important key. Third, using camera device for detecting the pothole [7][8][9] by the camera sensor. By the image processing with camera sensor has low in initial cost and fast detecting speed. Unfortunately, this method is sensitive to the outer environment such as light and rain.

We suggest automatic detecting pothole system using image processing with the camera sensor. This system concentrates in fast, real-time, high detecting accuracy and increasing the car suspensions life or others. To detect the pothole, we focus on three characteristics of the pothole. First, the rugged region of the pothole. Second, the pothole outlines have darker than outer region. Third, the pothole has the round shape.

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II. Pothole Detection

A. Proposed method

This System detects the pothole based on image processing. First, before to detecting process, we don't need RGB scale. Because the pothole region is dark, most of the asphalt pavement is gray and white. The system can detect the pothole with grayscale image. Fig. 1 shows after and before preprocessing which the RGB image and the grayscale image. It makes drop out unnecessary data.

This system is the parallel processing using superpixel, wavelet energy field and differential. The superpixel is clustering algorithm that clustering with the similar pixel. After finishing the clustering process, it makes the outlines of cells which are clustered pixels. The region of the pothole is rugged, the asphalt pavement is similarity region. So, we can segment the pothole from the outlines of superpixel's cells by checking polygonal of the superpixel's cells.

The wavelet energy field can show the texture of the image. So, it can be used checking the cell that is potholes or not. In Fig. 3 (b), if the cell includes the pothole, wavelet energy field reveals the potholes by highlighting the cell.

The differential method is based on similarity of asphalt pavement. The pothole, a road text and road lines are not continuous. Therefore, in Fig 3 (a), if we get the frame differential, it highlights the dissimilarity objects. To get more distinguishable highlighted image. Accumulating differential and subtract the first frame and second frame, then one more subtracts the second image with the result of the first and second differential. Detecting the pothole with three methods intersections.

B. Experiment

For testing the accuracy of the system, we used the asphalt

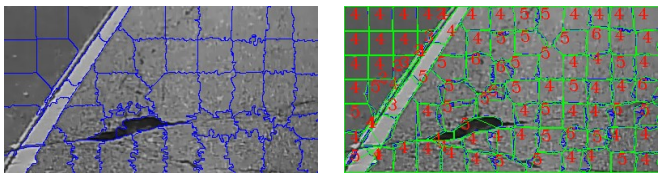
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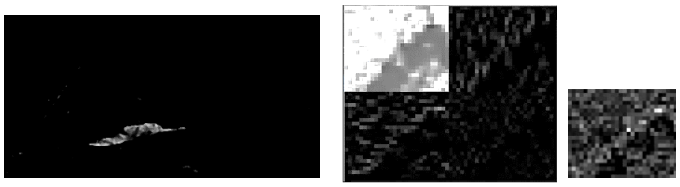
(a) (b)

Fig. 1 (a) Part of ROI image, (b) Gray scale of (a)



(a) (b)

Fig. 2 (a) Result of superpixel. It is show the outlines of cell. (b) Checking polygonal of outlines of cell



(a) (b)

Fig. 3 (a) Result of differential, (b) Result of wavelet transform and wavelet energy field

pavement video where the Korean highway. Then the camera was set on the front mirror. The camera device was Samsung Galaxy S7 and quality of the video was 1920 x 1080, 30 fps. The region of interest(ROI) was 800 x 200 pixels which included the asphalt pavement. The total recorded time is approximately 1minute 30 second equivalent to 38137 frames.

The video included 43 potholes. Testing system was running on windows 10 pro 64bit, i7-4550U (1.50GHz), 8 GB RAM, visual studio 2015 and OpenCV 3.2.

C. *Experiment result*

The result of experiment our system detected the 40 potholes in a total of 43 potholes. The miss detection, such as the asphalt pavement marking was 378 of total 1074 detection. Fig. 4 showed the success case of detecting the pothole. Fig. 5 showed failure in detecting potholes. While the system working the result, fps is 10 ~ 15 fps. Therefore, our system had 93.06% accuracy, 35% miss detection.

III. Conclusion And Future Work

We proposed automatic pothole detecting system based on image processing, using by wavelet, wavelet energy field and superpixel. We tested the system with the Korean highway asphalt pavement. The system's accuracy was 93.06%, miss detecting is 35%. Using the car's black-box can set our system and checking the location with GPS. The high -cost car has the automatic suspension system, but the cost is high and the general car doesn't have so our system can use commonly. We can expect improvement of the asphalt pavement condition, increase the efficiency of driving and the cars lifetime, protect of the car.



Fig. 4 Output image that detecting pothole and drawing the rectangle on pothole region



Fig. 5 Miss detection. Not a pothole but the pavement marking is dissimilarity. So, it makes miss detection

Next research will be making lower miss detection from the asphalt marking and porting the system in Android Operating System. Then, making automatic suspension system with using pothole detecting system and the map of the pothole.

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